

Ferroelectric FET based Signed Synapses of Excitatory and Inhibitory Connection for Stochastic Spiking Neural Network based Optimizer

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Abstract

For combinatorial optimization problem (CSP) solving of spiking neural networks (SNNs), both excitatory and inhibitory synaptic connections are necessary for mapping of constraints, along with adaptively-stochastic neuron. In this work, for the first time, a novel ferroelectric FET (FeFET) based signed synapse with only two transistors is proposed and experimentally demonstrated to achieve excitatory and inhibitory connections, enabling cascade circuit with our previous proposed FeFET-based adaptively-stochastic neuron for all ferroelectric SNN optimizer. Based on the proposed design, a stochastic SNN is implemented for fast solving CSPs with accuracy improvement by 200%, providing a promising ultralow-hardware-cost and energy-efficient solution for optimization.

(*Keywords:* Ferroelectric FET (FeFET), synapse, combinatorial optimization problems)

Introduction

Combinatorial optimization problems (CSPs) that require to find the best possible solution under certain constraints, are ubiquitous in energy-constrained edge AI applications [1-2] such as automotive logistics (Fig. 1a). Inspired by human brain, the spiking neural networks (SNNs) embedded with stochastic simulated annealing (SSA) algorithm [3] are promising to solve CSPs with high energy efficiency. For hardware implementation of SSA based SNN for CSPs (Fig.1b), signed synapse with function of both excitatory and inhibitory connection are necessary for mapping of constraints in CSPs, along with adaptively stochastic neuron for the convergence of SSA [2-4]. The emerging non-volatile memories, such as PCM [5] and RRAM [6], have been introduced to implement the signed synapse. However, they require complementary structure of 2T2R for signed behavior and complex neuron circuit for cascade, suffering from high hardware cost. Very recently, we have proposed and experimentally demonstrated a novel adaptively-stochastic spiking neuron with ultra-low hardware cost of only three transistors based on CMOS-compatible ferroelectric FET (FeFET) [7].

In this work, we further propose and experimentally implement an FeFET-based signed

synapse of both excitatory and inhibitory connection with only two transistors (Fig. 2a), which can be cascaded with our previously proposed adaptively-stochastic spiking neuron (Fig. 2b) for all FeFET based SNN to solve the CSPs with ultra-low hardware cost and high energy efficiency.

Design and experiment of FeFET-based signed synapse of excitatory and inhibitory connection

A. Signed synapse design based on FeFET

The constraints of CSP in SNN are mapped to excitatory and inhibitory synaptic connections between neurons, ensuring the convergence direction towards the optimal solution. As shown in Fig. 2a, in this work, the design of signed synapse consists of a PMOS (M1) and a FeFET (M2). The gate of PMOS is biased with a fixed voltage lower than the amplitude of input pulse. When the FeFET is programmed to the high threshold voltage (V_{TH}) state with the high channel resistance, the input pulses are applied to the neuron through PMOS, enabling excitatory connections. When FeFET is in the low V_{TH} state with lower channel resistance than the PMOS, the input port of adaptively-stochastic neuron will be pulled down to the negative voltage V_{inh} , enabling inhibitory connections.

B. FeFET-based signed synapse behavior

In our experiment, the 5-nm ferroelectric (FE) $Hf_{0.5}Zr_{0.5}O_2$ (HZO) film is adopted for the FeFET fabrication. When applying relatively large gate sweeping voltage, the I_d - V_g curves of FeFET show typical counterclockwise hysteresis (Fig. 3).

The proposed synapse circuit is further constructed, and signed synapse behavior is experimentally characterized. As shown in Fig. 4a, when the PMOS (M1) is turned on by the input signal and V_{g1} , with program/erase voltage applied to the gate of FeFET (M2) of plastic synapse, the synaptic output voltage can be set to excitation/inhibition and be maintained. When identical pulses are applied to gradually switch FE polarization, continuous modulation of synaptic output intensity can be achieved with good linearity and symmetry (Fig. 4b). As shown in Fig. 5, when receiving feedback spikes from the neurons, the synaptic output shows excitatory (positive) pulses with excitatory synaptic connection, while the synaptic output voltage keeps

negative when synapse sets inhibitory, indicating the signed connection for mapping of constraints in CSPs.

Cascade behavior of FeFET-based signed synapse and spiking neuron

In this work, to investigate the cascade behavior of proposed signed synapse and spiking neuron for all-FeFET SNN hardware system, the FeFET model including the polarization switching probability which follows a sigmoid dependence on the voltage amplitude [8] for each domain, is established based on dynamic Preisach ferroelectric polarization switching model [9] and BSIM-4 MOSFET model (Fig. 6). The simulated FE polarization switching processes of FE layers show increased variation of polarization switching dynamics when the applied voltage decreases or the domain number decreases with the smaller FE layer area (Fig. 6cd).

For the adaptively-stochastic neuron circuit in Fig. 2b, when receiving voltage pulse inputs, the voltage applied to soma leaky-FeFET ($V_{\text{den_soma}}$) will increase, as shown in the HSPICE circuit simulation results (Fig. 7). Accordingly, the stochasticity of spike generated by soma will adaptively decrease due to the increased applied voltage as shown in Fig. 7, indicating the decreasing stochasticity for SSA. Fig. 8 shows the cascade results of proposed FeFET-based signed synapse connection and adaptively-stochastic neuron, indicating that it can realize both excitatory and inhibitory constraints between neurons by programing/erasing FeFET (M2) of synapse, enabling all FeFET based SNN with SSA for the solution of CSPs.

Demonstration of SNN for CSP solving

Based on our proposed FeFET synapse and neuron design, the stochastic SNN enabling SSA is further studied for both optimization of travelling salesman problems (TSPs) and constraint satisfaction of Sudoku. For TSP abstracted from automotive logistics and scheduling systems, the SNN topology is two-dimensional, and for Sudoku, the SNN takes three-dimensional lattices with adaptively-stochastic neuron for each grid (Fig. 9a-b). The goals and rules are mapped to the signed synapses between neurons, and thus the adaptively stochastic SNN can gradually evolve from an initial state with annealing process. The results of solving process show that the systems can escape from the local minima and can finally converge to the optimal solution (Fig. 9c-d).

Moreover, the role of adaptively decreasing

stochasticity in SNN is studied, as shown in Fig. 9de. Taking Sudoku solving as an example, for SNN with neurons of fixed stochasticity, it will never converge to a stable solution, and only with adaptively decreased stochasticity can it escape from local minima and subsequently converge, along with 200% accuracy improvement. In addition, the proposed all FeFET-based SNN can significantly accelerate the solution of CSP by several orders of magnitude compared with traditional FPGA implementations and software approaches [10] (Fig. 10), showing its great potential for energy-efficient solution for CSPs.

Conclusion

This work proposed and experimentally demonstrated a novel FeFET based signed synapse of excitatory and inhibitory connections with only two transistors for the first time, enabling cascade circuit with the FeFET-based adaptively-stochastic neuron for all ferroelectric SNN optimizer for CSP solving with high area- and energy-efficiency.

Acknowledgments

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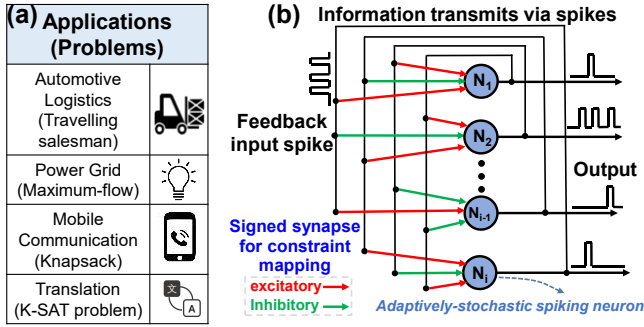


Fig. 1 (a) Edge applications of combinatorial optimization problems (CSPs) solving. (b) The schematic representation of brain-inspired spiking neural network (SNN) with signed synapse and adaptively-stochastic neuron for stochastic simulated annealing (SSA) based on recurrent connections.

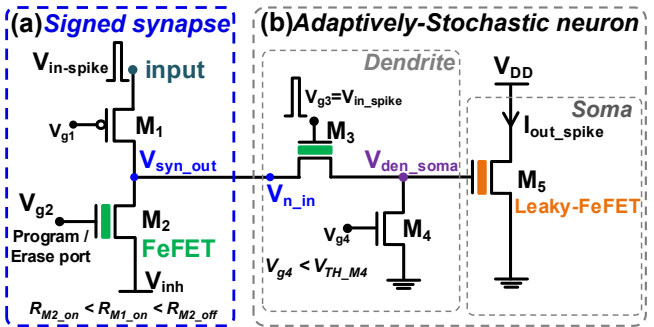


Fig. 2 The proposed FeFET-based (a) signed synapse and (b) adaptively-stochastic neuron. R_{M2_off}/R_{M2_on} is the channel resistance of FeFET (M2) after program (/erase). R_{M1_on} is the channel resistance of M1 with spike input. V_{inh} is the inhibitory negative voltage.

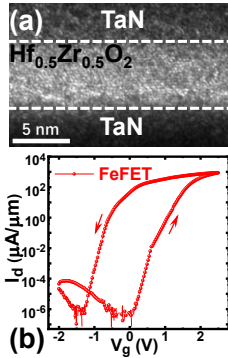


Fig. 3 Experimental characterization of FeFET (a) TEM image of FE layer. (b) I_D - V_G curves.

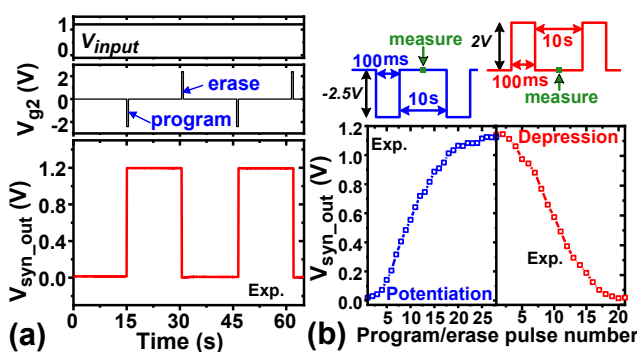


Fig. 4 Measured plastic behavior of signed synapse with an equivalent resistor used for M1. (a) The waveform of program/erase operation and the plastic output behavior. (b) The potentiation/depression characteristics in response to identical pulses.

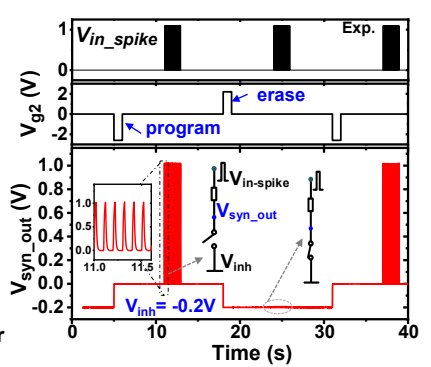


Fig. 5 Measured synaptic excitatory (/inhibitory) output with input pulses applied after program(/erase) of the signed synapse.

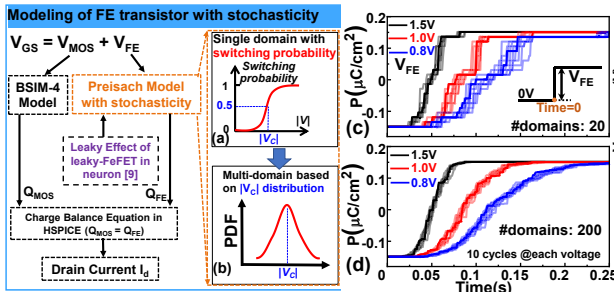


Fig. 6 The modeling framework of the ferroelectric transistor with stochasticity. (a) Single domain with voltage-dependent switching probability in (b) multi-domain Preisach model. (c) (d) Simulated FE polarization switching dynamics.

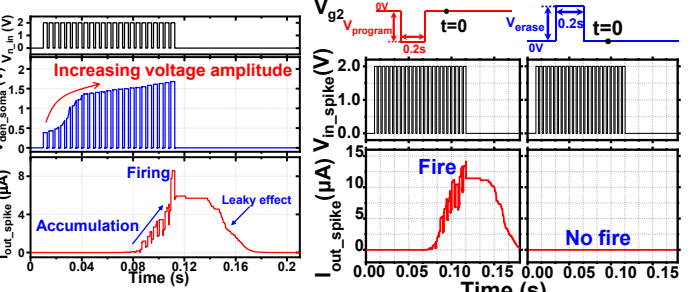


Fig. 7 Simulated proposed FeFET-based adaptively-stochastic neuron behavior.

Fig. 8 HSPICE simulated dynamics of proposed FeFET-based neuron cascaded with signed synapse.

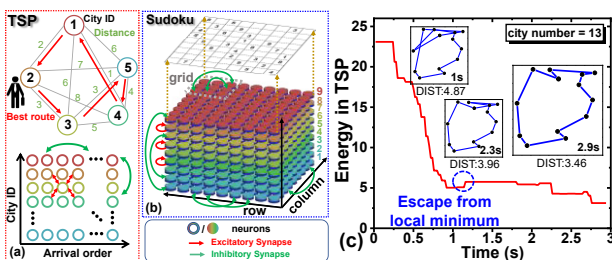


Fig. 9 The architecture of SNN based on proposed FeFET synapses and neurons for (a) traveling salesman problems (TSP) and (b) Sudoku solving by SSA. The constraint signed synaptic connections are mapped from the goals and rules of problems. (c) The solving process for TSPs. (d) The entropy and neuronal activities in the process of Sudoku solving. (e) The number of solved cases in the same period of time.

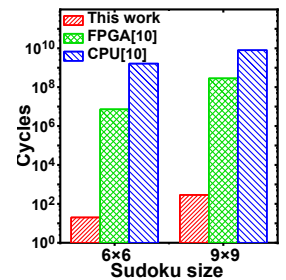


Fig. 10 The iteration cycles comparison for the solution of Sudoku based on different approaches.